

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

SMT. KUNDANBEN DINSHA PATEL DEPARTMENT OF INFORMATION TECHNOLOGY

Subject Name: Artificial Intelligence

Subject Code: IT473

Semester: B.Tech VII

Academic Year: 2019-20

Note: The laboratory will emphasize the use of Artificial Intelligence-Search, Game plying, Logic, NLP & Expert System Shells with Python Programming.

Instructions:

1. **All Practical must be performed individually and all experimental results must be uploaded on your respective EDMODO account.**
2. All Practical must be evaluated regularly in the laboratory by concern Lab Teacher.
3. Each practical answer would be evaluated as learning outcome.

Practical List

Sr. No.	Aim of the Practical	Hrs	COs	POs	PEOs
Pre Req.	Introduction to Prolog and History, Python Programming. Installation & Configuration of Python. Along with its all major editors, IDLE, Anaconda, Jupyter, Interpreter, etc.	-	-	-	-
1.	The 8 puzzle consists of eight numbered, movable tiles set in a 3x3 frame. One cell of the frame is always empty thus making it possible to move an adjacent numbered tile into the empty cell. Apply intelligence to solve given problem and Implement it.	2	1,5	2,6,7	1,2
2.	Implement Uninformed Search Strategies: BFS, DFS, and ID (Iterative Deepening-Idea of Game Searching (Chess, Checkers, Tic-Tac-Toe, Puzzle Game etc.))	4	1,5	2,6,7	1,2
3.	Implement Informed (heuristic) search strategy for optimal shortest path and traffic navigational system. (Hint: A* algorithm by mentioning OPEN and CLOSED List and it takes a heuristic function as Euclidian Distance Equation) Implement Informed (heuristic) search strategy for problem reduction and for planning good algorithms. (Hint: AO*-AND/OR graph by mentioning OPEN and CLOSED List).	4	1,5	2,6,7	1,2
4.	Getting Started With Python Logic Programming using Kanren and SymPy packages • Compare expressions and find out unknown values. • Form and manipulate logic expressions using symbolic and Boolean values with SymPy package.	4	2	5,6,7	2,3
5.	Implement Game Search algorithm that seeks to decrease the number of nodes that are evaluated by the minimax algorithm in its search tree. It is an adversarial search algorithm used commonly for machine playing of two-	2	2	5,6,7	2,3

	player games (Tic-tac-toe, Chess, Go, etc.). It stops evaluating a move when at least one possibility has been found that proves the move to be worse than a previously examined move. Such moves need not be evaluated further.				
6.	Understand Text Summarization and Implement your own summarizer in python.	2	2	5,6,7	2,3
7.	Implement Bayes rule with probabilistic inference for reasoning under uncertainty.	2	3	1	1,5
8.	Implement Deep Q- Learning (Deep Q-Network) in Reinforcement Learning (RL) using Python Question: Compare Deep RL vs. Deep Learning	2	3	1	1,5
9.	Implement following Computer Vision task on Images using OpenCV.	4	3	1	1,5
	Reading, Writing and Displaying Images				
	Image Filtering				
	Changing Color Spaces				
	Image Contours				
	Image Rotation				
	Scale Invariant Feature Transform				
	Image Translation				
	Speeded-Up Robust Features (SURF)				
	Image Segmentation (Watershed Algorithm)				
	Feature Matching				
	Edge Detection				
	Face Detection				
10	Kaggle machine learning challenges: - Predicting Molecular Properties - Detect objects in varied and complex images - Detect pairs of objects in particular relationships - Learn computer vision fundamentals with the famous MNIST data - Predict survival on the Titanic and get familiar with ML basics - Predict sales prices and practice feature engineering, RFs, and gradient boosting - ImageNet Object Localization Challenge. Identify the objects in images - Predict Future Sales	4	2,3	1	1,5
11	Case Study1: Natural language processing and Expert System. Case Study2: Briefly mention any three artificial intelligence libraries in Python	-	2,3	1	1,5